

Problem 2

Problem 1 Determine the following information about the given function and then graph the function:

$$f(x) = \frac{x^2 + x + 1}{x^2}$$

Domain

x, y -intercepts

Symmetry

Asymptotes

Intervals of increasing/decreasing

Maxima/Minima

Intervals of concavity

Inflection points

Explanation.

(a) **Domain**

The function is a rational function, and so the domain of the function is all real numbers except where the denominator equals zero.

$$x^2 = 0 \implies x = 0$$

So the domain of f is $(-\infty, 0) \cup (0, \infty)$.

(b) **x, y -intercepts**

To find any x -intercept(s), set $y = 0$ and solve:

$$\begin{aligned} \frac{x^2 + x + 1}{x^2} = 0 &\implies x^2 + x + 1 = 0 \\ &\implies x = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2(1)} \end{aligned}$$

which has no real solutions. Thus, f has no x -intercepts.

Since $x = 0$ is not in the domain of f , f has no y -intercepts as well.

(c) **Symmetry**

Note that $f(1) = 3$ and $f(-1) = 1$. So it cannot be for all values of x that either $f(-x) = f(x)$ or $f(-x) = -f(x)$. So f is neither even nor odd, and therefore f has no symmetry.

(d) **Asymptotes**

Vertical Asymptotes: Our only candidate is $x = 0$, and so we compute the two one-sided limits:

$$\lim_{x \rightarrow 0^-} \frac{x^2 + x + 1}{x^2} = \infty$$

$$\lim_{x \rightarrow 0^+} \frac{x^2 + x + 1}{x^2} = \infty$$

Therefore, $x = 0$ is the only vertical asymptote of f .

Horizontal Asymptotes: We compute the following limits:

$$\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^2} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 + x + 1}{x^2} = 1$$

we checked both ends and so the only horizontal asymptote of f is $y = 1$.

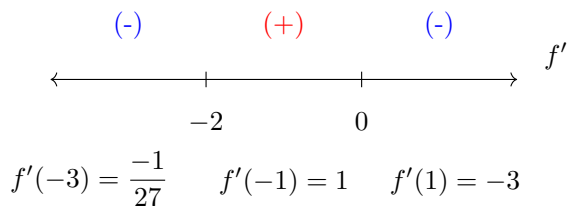
(e) **Increasing/Decreasing**

$$\begin{aligned} f'(x) &= \frac{x^2(2x+1) - (x^2+x+1)(2x)}{x^4} \\ &= \frac{2x^3 + x^2 - 2x^3 - 2x^2 - 2x}{x^4} \\ &= \frac{-x^2 - 2x}{x^4} \\ &= \frac{-x - 2}{x^3} \end{aligned}$$

To find where f' is positive and where f' is negative, we need to find where $f'(x) = 0$ and where $f'(x)$ does not exist. Clearly, $f'(x)$ does not exist when $x = 0$. To find when $f'(x) = 0$, we solve:

$$\begin{aligned} \frac{-x-2}{x^3} = 0 &\implies -x-2 = 0 \\ &\implies -x = 2 \\ &\implies x = -2 \end{aligned}$$

Since $x = 0$ is not in the domain of f , $x = -2$ is the only critical point of f . To see where f is increasing and decreasing, consider the following sign chart for f' :



So we see that f is increasing on $(-2, 0)$, and f is decreasing on $(-\infty, -2)$ and $(0, \infty)$.

(f) **Local Extrema**

f' changes from negative to positive at $x = -2$, so this is the location of a local minimum. f' also changes from positive to negative at $x = 0$, but f is not defined at $x = 0$ and so this is not a local extreme value. f has a local minimum at $\left(-2, \frac{3}{4}\right)$.

(g) **Concavity**

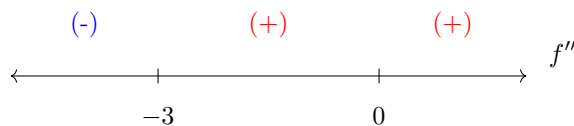
$$\begin{aligned} f''(x) &= \frac{x^3(-1) - (-x-2)(3x^2)}{x^6} \\ &= \frac{-x^3 + 3x^3 + 6x^2}{x^6} \\ &= \frac{2x^3 + 6x^2}{x^6} \\ &= \frac{2x + 6}{x^4} \\ &= \frac{2(x+3)}{x^4} \end{aligned}$$

To find where f'' is positive and where f'' is negative, we need to find where $f''(x) = 0$ and where $f''(x)$ does not exist. Clearly, $f''(x)$ does not exist when $x = 0$. To find when $f''(x) = 0$, we solve:

$$\begin{aligned} \frac{2(x+3)}{x^4} = 0 &\implies 2(x+3) = 0 \\ &\implies x = -3 \end{aligned}$$

The denominator of f'' is always positive so the sign of f'' depends on the numerator. When $x < -3$, $f'' < 0$ and when $x > -3$, $f'' > 0$.

To see where f is concave up and concave down, consider the following sign chart for f'' :



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So we see that f is concave up on $(-3, 0)$ and $(0, \infty)$, and f is concave down on $(-\infty, -3)$.

(h) **Inflection Points**

$f''(x)$ changes sign from negative to positive at $x = -3$ so f has an inflection point at $\left(-3, \frac{7}{9}\right)$

(i) **The graph of f**

